Lab: Build the OrbitCo Operational Dashboard

Type: Hands-on lab (individual work) **Time:** 25 minutes (20 min individual work + 5 min share-out)

Objective

Starting from the cleaned workbook produced in Lesson 1, build a one-page operational dashboard that combines Key Performance Indicator (KPI) tiles, a pivot-driven trend chart, a severity-mix composition chart, and a conditionally-formatted details band. The finished dashboard is the visual artifact you will embed in the capstone briefing deck in Lesson 4, and it is the single best demonstration of the On-the-Job Learning (OJL) 5d competency in the course.

Scenario

You are the operations analyst at OrbitCo and your director has asked for the 30-day review view. The `Tickets` Table (413 rows), `Uptime` Table, `SLALookup` Table, and `outputs` KPI summary from the Lesson 1 labs are all in place. Now the director wants "one page that tells me how the quarter is going — tiles at the top, charts in the middle, the row-level detail underneath if I want to look." You have 20 minutes to build exactly that.

Instructions

1. Open your `orbitco-workbook-[yourname].xlsx` from Lesson 1 (the file with the `Tickets`, `Uptime`, `SLALookup`, and `outputs` sheets). Insert a new sheet and name it `dashboard`. This is the only sheet the director will look at; the rest are plumbing.
2. At the top of the `dashboard` sheet, build a strip of five KPI tiles. For each tile, use two cells — a small label cell above and a large value cell below. Reference the `outputs` sheet (or use a direct structured-reference formula) for each value:
   - **Total tickets (30d):** `=COUNTA(Tickets[TicketID])`
   - **MTTR, minutes:** `=ROUND(AVERAGE(Tickets[ResolutionMinutes]),0)`
   - **High-severity tickets:** `=COUNTIFS(Tickets[Severity],"High")`

- **Service Level Agreement (SLA) breaches:** `=COUNTIFS(Tickets[SLAStatus],"BREACH")`
- **Uptime %:** `=1 - SUM(Uptime[DowntimeMinutes]) / (30*24*60)` — format as percentage with one decimal place

3. Build a helper column on the `tickets_clean` sheet named `OpenedWeek` with the formula `=WEEKNUM([@OpenedAt])` so the pivot has a week field. On a new sheet named `pivots`, insert a PivotTable from the `Tickets` Table. Drag `OpenedWeek` to Rows, `TicketID` to Values (Count). Insert a **PivotChart → Line** of this pivot and then copy the chart onto the `dashboard` sheet in the middle band. This is the "tickets per week" trend.
4. Insert a sparkline next to the MTTR tile: **Insert → Sparkline → Line**, source range is a daily-MTTR helper column on the `tickets_clean` sheet (one row per date, `=AVERAGEIFS(Tickets[ResolutionMinutes], Tickets[OpenedDate], [@OpenedDate])`). The sparkline lives inside the single cell next to the MTTR value.
5. On the `pivots` sheet, insert a second PivotTable: Rows = `Severity`, Values = Count of `TicketID`. Open **Value Field Settings → Show Values As → % of Grand Total**. Insert a **PivotChart → Doughnut** and copy it onto the `dashboard` sheet next to the line chart. This is the severity-mix donut.
6. Apply an icon-set traffic-light rule on the Uptime % tile: **Home → Conditional Formatting → Icon Sets → 3 Traffic Lights**. In the rule manager, set thresholds to green ≥ 0.999, yellow ≥ 0.995, otherwise red. Because the cell value is a decimal, the thresholds must also be decimals — `0.999`, not `99.9`.
7. Below the charts on the `dashboard` sheet, paste a copy of the `Tickets` Table values (or use a formula-driven reference if you prefer — either approach is acceptable for this lab). Apply two conditional-formatting rules: red fill where `SLAStatus = "BREACH"` and bold where `Severity = "High"`. Make sure the rules apply to the Table column, not a fixed range.
8. Add a freshness timestamp in the top-right corner of the `dashboard` sheet: a label cell reading `Data as of:` and an adjacent cell with the formula `=TODAY()`.
9. Add a `README` sheet at the far left of the workbook. Fill it with five short sections: purpose, data source, refresh steps, KPI definitions, and owner. This is the OJL 5d artifact inside the artifact — a dashboard without a `README` is a dashboard nobody can maintain after you leave.

10. On the `dashboard` sheet, **View → uncheck Gridlines** for a cleaner presentation feel. Save the file as `orbitco-dashboard-[yourname].xlsx`.

Example

A well-built `dashboard` sheet reads, top-to-bottom:

Region	Content
Top-right corner	`Data as of: 4/13/2026`
Top band (tiles)	`Total Tickets 413` · `MTTR 47 min` (with a 30-day sparkline beside it) · `High 62` · `SLA Breaches 14` · `Uptime 99.92%` (with a green traffic-light icon)
Middle band (charts)	Line chart of `Tickets per Week` on the left; Doughnut of `Severity Mix %` on the right
Bottom band (details)	A `Tickets` Table replica with red fills on 14 rows (BREACH) and bold on 62 rows (High)

No raw data is visible on this sheet. The `Tickets` and `Uptime` Tables still live on their own tabs; the pivots live on a hidden `pivots` sheet; the `README` is one click away.

Expected Deliverables

- A single `.xlsx` file named `orbitco-dashboard-[yourname].xlsx`.
- A `dashboard` sheet with five KPI tiles, one sparkline, two pivot-driven charts (line for weekly trend, doughnut for severity mix), a conditionally-formatted details band, a `Data as of:` timestamp, and no visible gridlines.
- A `pivots` sheet holding the two PivotTables.
- A `README` sheet with five sections (purpose, data source, refresh steps, KPI definitions, owner).
- An icon-set traffic-light rule on the Uptime % tile with thresholds set as decimals (0.999 / 0.995).

- Conditional formatting on the details band applied to the Table column so new rows inherit the rule.

Success Criteria

You have completed this lab successfully when you can:

- Open your workbook to the `dashboard` tab and have every number, chart, and icon render correctly without switching to another sheet to investigate.
- Point to any KPI tile and trace the formula back to a structured reference on the source Table (no hardcoded `=Sheet2!D428` cell addresses).
- Add a row of synthetic test data to the `Tickets` Table, right-click a pivot on the `pivots` sheet, choose **Refresh**, and confirm the line chart, the donut, and the KPI tiles all update consistently.
- Explain in one sentence why the traffic-light thresholds are set at 0.999 and 0.995 instead of 99.9 and 99.5.
- Hand the workbook to a colleague who has never seen it and have them find the data source, refresh procedure, and KPI definitions from the `README` tab in under a minute.